

Sentiment Analysis Using Machine Learning

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Bachelor of Engineering

by

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CERTIFICATE

This Mini Project report entitled “**Sentiment Analysis using Machine Learning**” by **Mr. Rohit Bhamare, Mr. Zuhaib Sayyed, Ms . Shubhangini Paithankar , Ms. Spurti Kamble** is approved for the degree of **Bachelor of Engineering in Information Technology** for academic year 2023 – 2024.

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Declaration

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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LIST OF ABBREVIATION

ABBREVIATIONS	DEFINITION
UML	UNIFIED MODELING LANGUAGE
E-R	ENTITY – RELATIONSHIP
SA	SENTIMENTAL ANALYSIS
IDE	INTEGRATE DEVELOPMENT
NLP	NATIONAL LANGUAGE PROCESS
WBS	WORK BREAKDOWN STRUCTURE

ABSTRACT

Sentiment analysis (SA) is a process of extensive exploration of data stored on the Web to identify and categorize the views expressed in a part of the text. The intended outcome of this process is to assess the author attitude toward a particular topic, movie, product, etc. The result is positive, negative, or neutral. These study illustrated different techniques in SA approach for extracting and analytics sentiments associated with the polarity of positive, negative, or neutral on the topic selected. Social networks SA can be a useful source of information and data. SA acquires important in many areas of business, politics, and thought. So, this study contains a comprehensive overview of the most important studies in this field from the past to the recent studies till 2017. The main aim of this study is to provide full concept about SA techniques and its classification and methods used it. Also, we give a brief overview of big data techniques and its relation and use in SA field. Because the recent period has witnessed a remarkable development in the use of Big Data (Hadoop) in the process collection of data and reviews from social networks for analysis.

Keywords : Big data, Classification, Challenges, Sentiment analysis, Social media, twitter, amazon.

CHAPTER 1

INTRODUCTION

1. INTRODUCTION

1.1 Motivation :

Sentiment analysis involves the use of data mining, machine learning (ML), artificial intelligence and computational linguistics to mine text for sentiment and subjective information such as whether it is expressing positive, negative or neutral feelings.

1.2 Problem statement

The term text analytics describes a set of linguistic, statistical, and machine learning techniques that model and structure the information content of textual sources for business intelligence, exploratory data analysis, research, or investigation. The term is roughly synonymous with text mining; indeed, Ronen Feldman modified a 2000 description of "text mining" in 2004 to describe "text analytics". The latter term is now used more frequently in business settings while "text mining" is used in some of the earliest application areas, dating to the 1980s, notably life-sciences research and government intelligence. The term text analytics also describes that application of text analytics to respond to business problems, whether independently or in conjunction with query and analysis of fielded, numerical data. It is a truism that 80 percent of business-relevant information originates in unstructured form, primarily text. These techniques and processes discover and present knowledge: facts, business rules, and relationships - that is otherwise locked in textual form, impenetrable to: automated processing.

1.3. Objective

- With the explosion of different opinions through social networking sites, new ideas were generated by systems, politicians, psychologists, manufacturers, and researchers to analyze them to implement the best decisions.
- Sentiment analysis has a high efficiency using NLP, as statistics, and machine learning approaches to extract and define sentiment content in a text unit.

1.4 Existing System :

Existing approaches to sentiment analysis can be grouped into three main categories knowledge-based techniques, statistical methods, and hybrid approaches. Knowledge-based techniques classify text by affect categories based on the presence of unambiguous affect words such as happy, sad, afraid, and bored. Some knowledge bases not only list obvious affect words, but also assign arbitrary words a probable "affinity to particular emotions. Statistical methods leverage elements from machine learning such as latent semantic analysis, support vector machines, "bag of words". "Point wise Mutual Information" for Semantic Orientation, and deep learning. More sophisticated methods try to detect the holder of a sentiment and the target. To mine the opinion in context and get the feature about which the speaker has opined, the grammatical relationships of words are used. Grammatical dependency relations are obtained by deep parsing of the text. Hybrid approaches leverage both machine learning and elements from knowledge representation such as ontologies and semantic networks in order to detect semantics that are expressed in a subtle manner, eg, through the analysis of concepts that do not explicitly convey relevant information, but which are implicitly linked to other concepts that do so.

1.5 Scope :

Sentiment analysis is simply the process categorizing the sentiments underlying a text. It is such a simple task that it can also be done manually; simply read each piece of feedback and determine whether it is positive or negative of Among many analytical fields, one in which humans outperform all others is the ability to recognize feelings. However, for feedback presented to you, such as 40-50 or even 100, this is doable. However, if you have a data set of, say, 10,000 reviews, manually analyzing them becomes impossible.

CHAPTER 2

LITERATURE SURVEY

2. LITERATURE SURVEY

Shihab Elbagir , Nov 2019

In recent years, research on Twitter sentiment analysis, which analyzes Twitter data (tweets) to extract user sentiments about a topic, has grown rapidly. Many researchers prefer the use of machine learning algorithms for such analysis. This study aims to perform a detailed sentiment analysis of tweets based on ordinal regression using machine learning techniques.

Jianming Zhang , July 2019

Deeper neural networks have achieved great results in the field of computer vision and have been successfully applied to tasks such as traffic sign recognition. However, as traffic sign recognition systems are often deployed in resource-constrained environments, it is critical for the network design to be slim and accurate in these instances.

Yan Li , July 2022

This paper presents an in-depth study of the sentiment of social network communication through a deep learning-based natural language processing approach and designs a corresponding model to be applied in the actual social process. Specifically, the network can dynamically select the most important word in the current state according to the information available and achieve the accurate recognition of the dynamically changing important content in a sentence.

Amir Leshem , April 2017

Opinion dynamics have fascinated researchers for centuries. The ability of societies to learn as well as the emergence of irrational {\it herding} are equally evident. The simplest example is that of agents that have to determine a binary action, under peer pressure coming from the decisions observed. By modifying several popular models for opinion dynamics so that agents internalize actions rather than smooth estimates of what other people think, we are able to prove that almost surely the actions final outcome remains random, even though actions can be consensual or polarized depending on the model

John R. P. French , May 1956

This theory illustrates a way by which many complex phenomena about groups can be deduced from a few simple postulates about interpersonal relations. By the application of digraph theory we are able to treat in detail the patterns of relations whose importance has long been noted by the field theorists." Three major postulates are presented as well as a variety of theorems dealing with the effects of the power structure of the group, the effects of communication patterns, the effects of patterns of opinion, and leadership. 32 references. (PsycINFO Database Record (c) 2006 APA, all rights reserved).

Table 2.1 Literature Survey Table

Sr. no.	Author Name	Topic Name	Year of Publication	Working / Purpose	Technology used
1	Shihab Elbagir	Twitter Sentiment Analysis Based on Ordinal Regression	2019	to perform a detailed sentiment analysis of tweets based on ordinal regression using machine learning techniques	NLP
2	Jianming Zhang	Lightweight deep network for traffic sign classification	2019	Deeper neural networks have achieved great results in the field of computer vision and have been successfully applied to tasks such as traffic sign recognition.	Python , Open CV
3	Yan Li	Fuzzy Rule based Unsupervised Sentiment Analysis from Social Media Posts	2022	in-depth study of the sentiment of social network communication through a deep learning-based natural language processing approach and designs a corresponding model to be applied in the actual social process	NLP
4	Amir Leshem	Tourism Mobile App With Aspect-Based Sentiment Classification Framework for Tourist Reviews	2017	The ability of societies to learn as well as the emergence of irrational are equally evident	Python
5	John R. P. French	Telugu Tweets Sentiment Analysis Based on Ordinal Regression	1956	we are able to treat in detail the patterns of relations whose importance has long been noted by the field theorists	Python

CHAPTER 3

RESEARCH GAP

3. RESEARCH GAP

Knowledge-based techniques classify text by affect categories based on the presence of unambiguous affect words such as happy, sad, afraid, and bored. Some knowledge bases not only list obvious affect words, but also assign arbitrary words a probable "affinity" to particular emotions. Statistical methods leverage elements from machine learning such as latent semantic analysis, support vector machines, "bag of words", "Point wise Mutual Information" for Semantic Orientation, and deep learning. More sophisticated methods try to detect the holder of a sentiment and the target. To mine the opinion in context and get the feature about which the speaker has opined, the grammatical relationships of words are used. Grammatical dependency relations are obtained by deep parsing of the text. Hybrid approaches leverage both machine learning and elements from knowledge representation such as ontologies and semantic networks in order to detect semantics that are expressed in a subtle manner, c.g., through the analysis of concepts that do not explicitly convey relevant information, but which are implicitly linked to other concepts that do so

Sentiment analysis may involve analysis of movie reviews for estimating how favorable a review is for a movie. Such an analysis may need a labeled data set or labeling of the affectivity of words. Resources for affectivity of words and concepts have been made for WordNet and ConceptNet, respectively. Text has been used to detect emotions in the related area of affective computing. Text based approaches to affective computing have been used on multiple corpora such as students evaluations, children stories and news stories..

CHAPTER 4

RESEARCH OBJECTIVE

4. RESEARCH OBJECTIVE

- The term text analytics describes a set of linguistic, statistical, and machine learning techniques that model and structure the information content of textual sources for business intelligence, exploratory data analysis, research, or investigation.
- The term is roughly synonymous with text mining; indeed.
- Ronen Feldman modified a 2000 description of "text mining" in 2004 to describe "text analytics".
- The latter term is now used more frequently in business settings while "text mining" is used in some of the earliest application areas, dating to the 1980s, notably life-sciences research and government intelligence.
- The term text analytics also describes that application of text analytics to respond to business problems, whether independently or in conjunction with query and analysis of fielded, numerical data.
- It is a truism that 80 percent of business-relevant information originates in unstructured form, primarily text.
- These techniques and processes discover and present knowledge -facts, business rules, and relationships - that is otherwise locked in textual form, impenetrable to automated processing.

CHAPTER 5

PROPOSED SYSTEM

5. PROPOSED SYSTEM

Sentiment analysis is the use of natural language processing, text analysis, computational linguistics, and biometrics to systematically identify, extract, quantify, and study affective states and subjective information. Sentiment analysis is widely applied to voice of the customer materials such as reviews and survey responses, online and social media, and healthcare materials for applications that range from marketing to customer service to clinical medicine. With the rise of deep language models, such as ROBERT a, also more difficult data domains can be analyzed, eg. News texts where authors typically express their opinion/sentiment less explicitly.

5.1. SYSTEM ARCHITECTURE

A system architecture is the conceptual model that defines the structure, behaviour, and more views of a system. An architecture description is a formal description and representation of a system. Organized in a way that supports reasoning about the structures and behaviours of the system. System architecture conveys the informational content of the elements consisting of a system, the relationships among those elements, and the rules governing those relationships. The architectural components and set of relationships between these components that an architecture description may consist of hardware, software, documentation, facilities, manual procedures, or roles played by organizations or people.

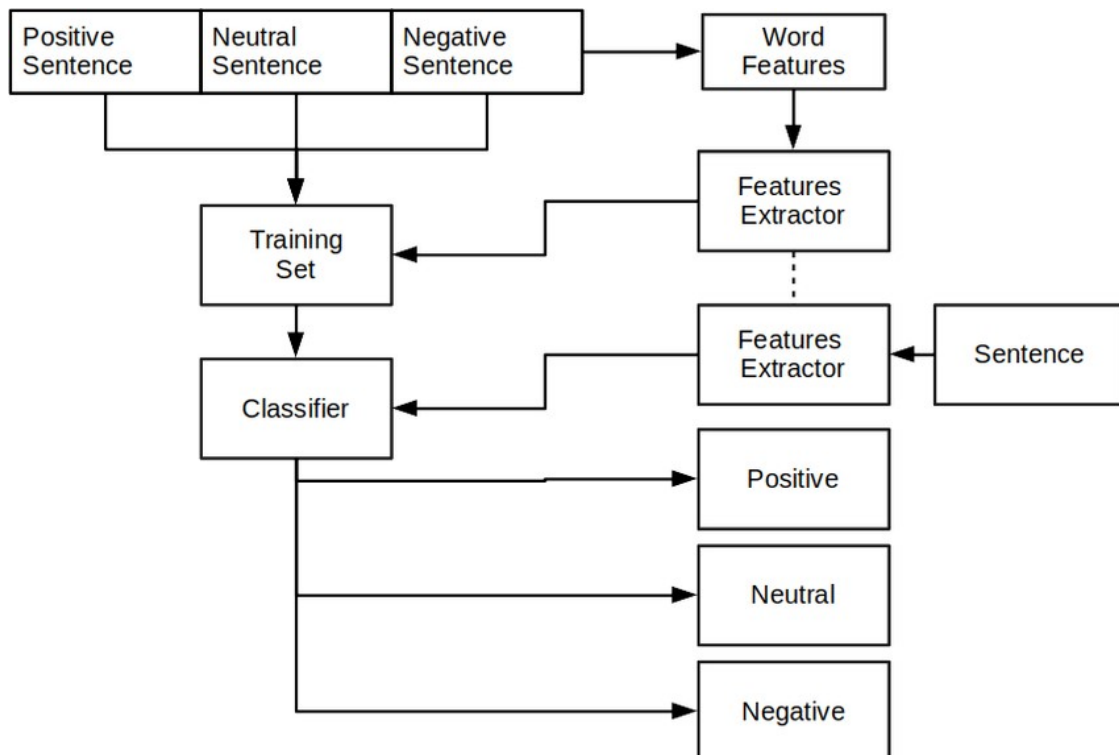


Fig. : 5.1 System Architecture Diagram

CHAPTER 6

SYSTEM REQUIREMENT

6. SYSTEM REQUIREMENT

6.1] Software Requirement :

- Platform Used: Windows 10
- Anaconda Distribution
- Python 3.x
- NLTK Toolkit
- Sklearn Requests
- Toolkit vaderSentiment
- Flask

6.2] Hardware Requirement :

- Laptop or Computer
- Processor: Intel core i3
- RAM: 8GB
- Storage: 500MB to 1TB
- Internet Connection Required

6.3] Technology used :

- Python
- NLP
- HTML
- CSS
- JAVASCRIPT

CHAPTER 7

SYSTEM DESIGN

7.1 UML diagram :

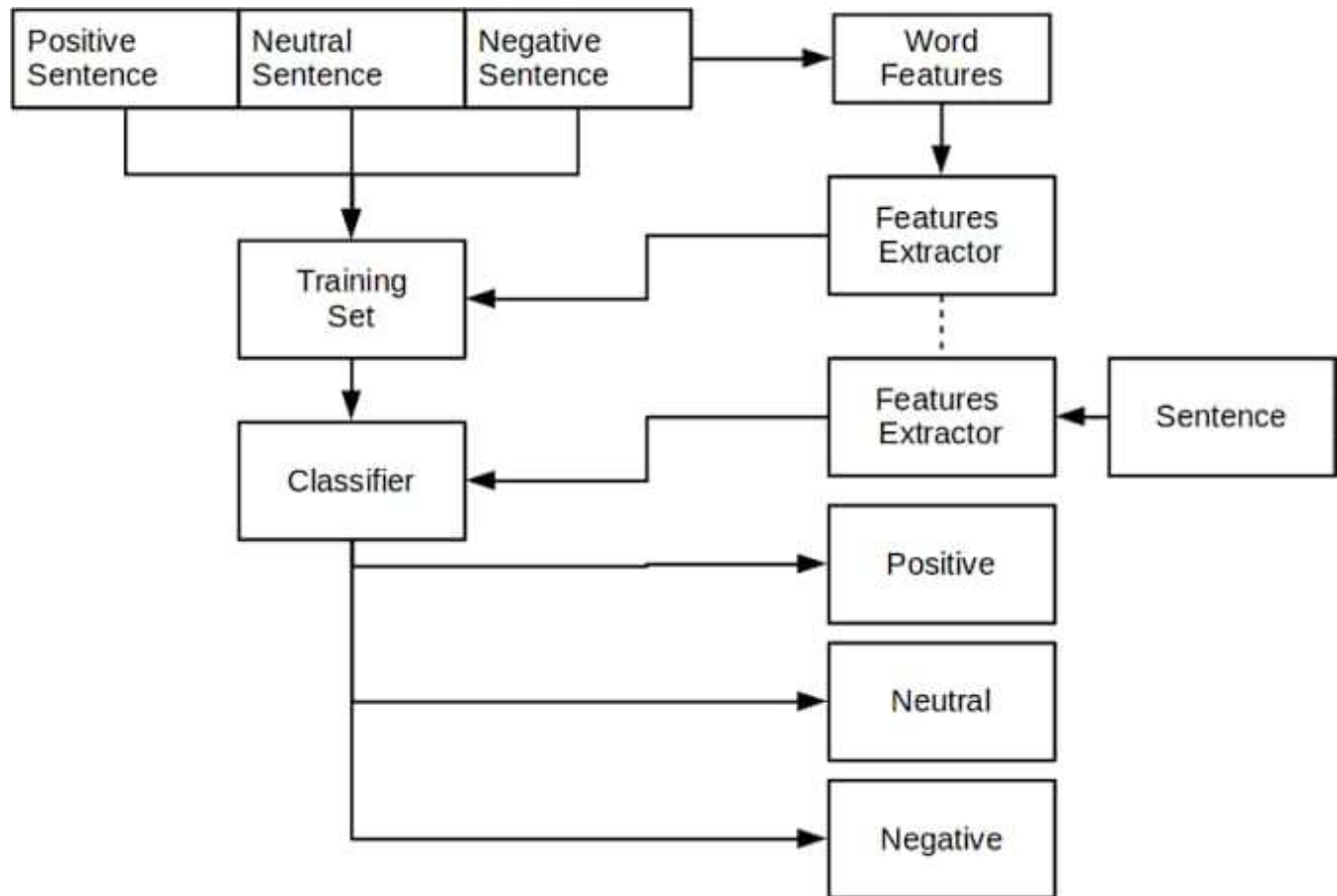


Fig. : 7.1 UML diagram

7.2 DFD diagram

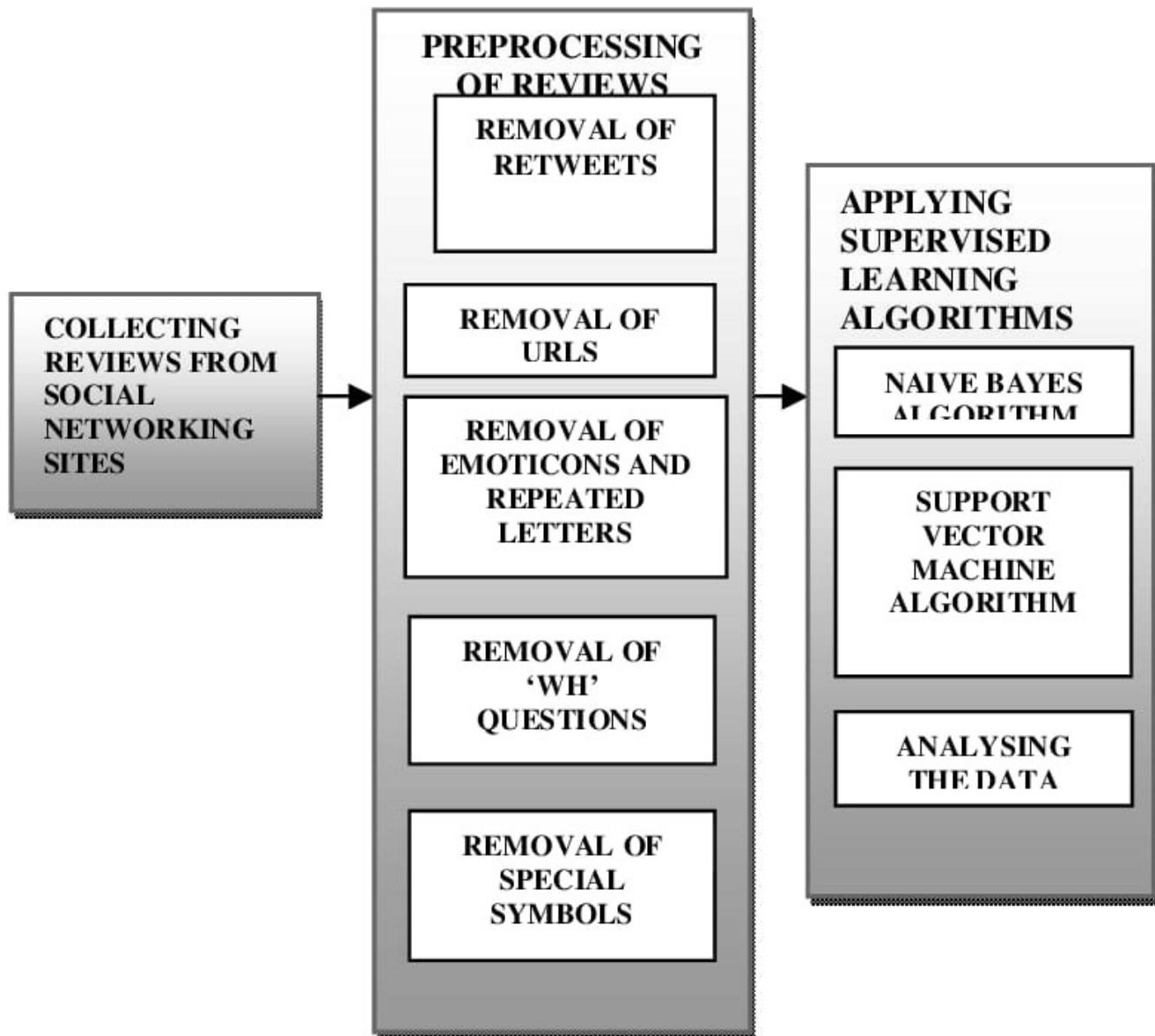


Fig. : 7.2 DFD diagram

CHAPTER 8

PROJECT IMPLEMENTATION

8. RESULT AND IMPLEMENTATION

8.1 Result / Output

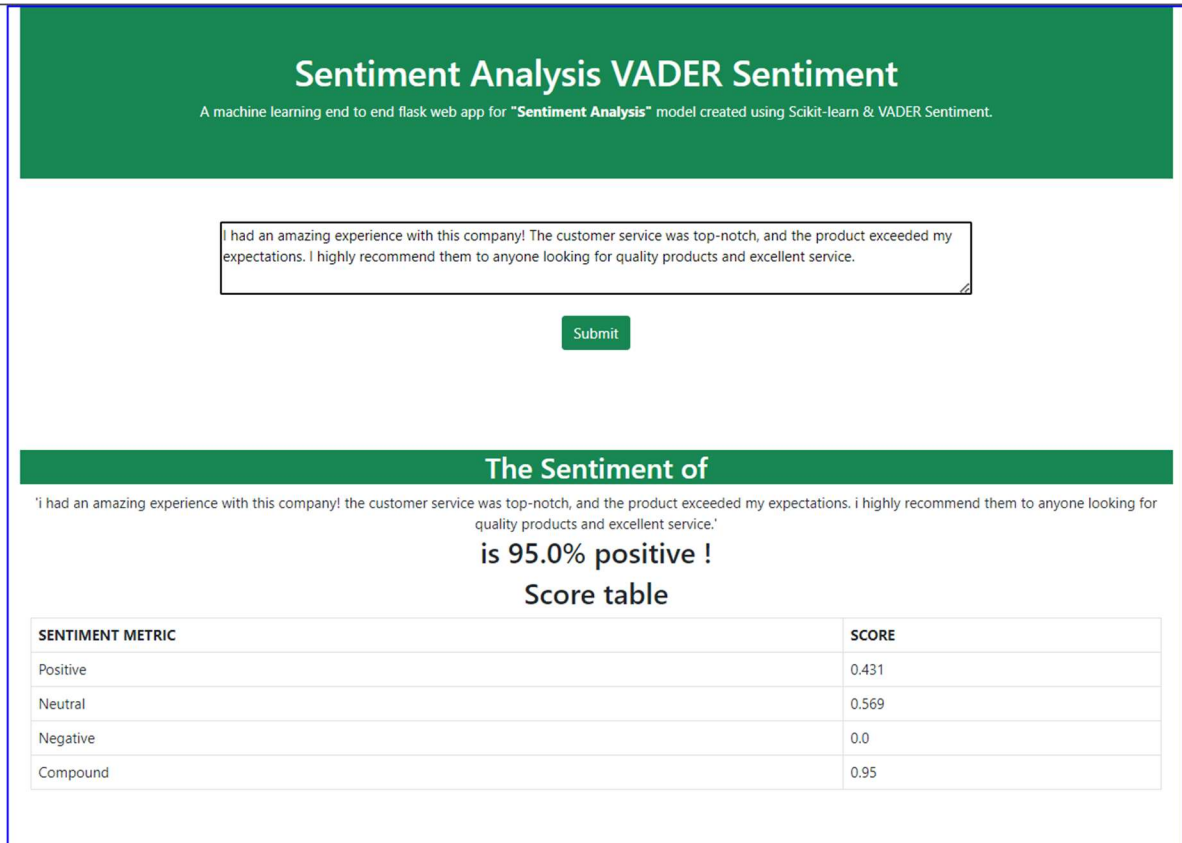


Fig. : 8.1 Positive review

Here In this figure 8.1 , we have taken a screen shot of our web application in which we have perform a example of a Positive review . Here we have taken a review of one customer who bought a product of some company , And we check that review in our Sentiment Analysis so we got a score of 95.0 % Positive.

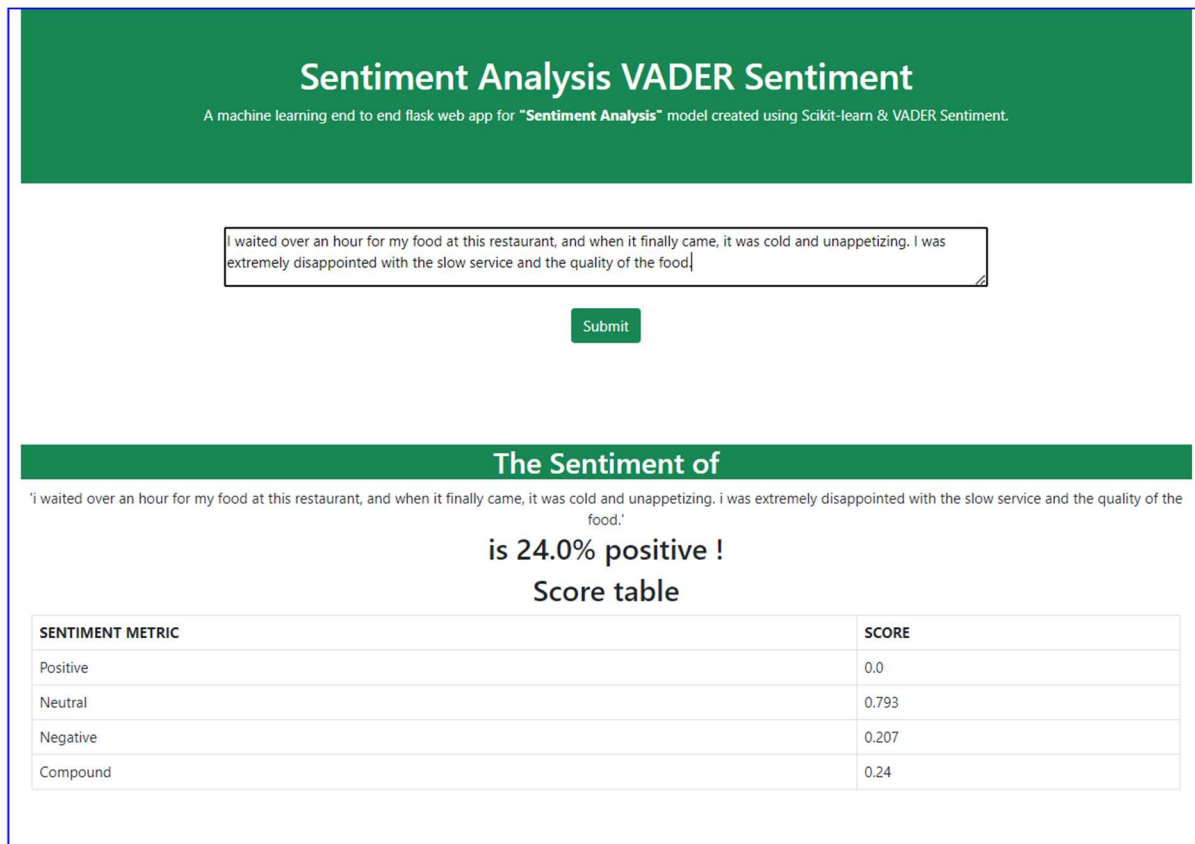


Fig. : 8.2 Negative Review

Here In this figure 8.2 , we have taken a screen shot of our web application in which we have perform a example of a Negative review . Here we have taken a review of one customer who order a food at some restaurant , And we check that review in our Sentiment Analysis so we got a score of 24.0 % Positive. 24 % is very low positive review that means it is negative .

Sentiment Analysis VADER Sentiment

A machine learning end to end flask web app for "Sentiment Analysis" model created using Scikit-learn & VADER Sentiment.

Submit

The Sentiment of

'this place is okay'

is 61.0% positive !

Score table

SENTIMENT METRIC	SCORE
Positive	0.655
Neutral	0.345
Negative	0.0
Compound	0.61

Fig. : 8.3 Neutral review

Here In this figure 8.3 , we have taken a screen shot of our web application in which we have perform a example of a okay review (neutral review). Here we have taken a review of one visitor who visit a garden and we just gave a simple review that “ This place is okay ”, this means not bad also not very good , And we check that review in our Sentiment Analysis so we got a score of 61.0 % Positive. That means it is okay. It is little more than 50% so this review will be consider as neutral review.

CHAPTER 9

CONCLUSION

CONCLUSION

The main advantage and feature of the project is that its simplicity attracts a lot of users. Login and registration of the employee are done in this application for security purposes and to identify the employees. Admin can keep track of task information. If the task is complete then the admin can see the employee current location on Google map, Also, the admin can see can task completed photo for a particular task. If the task is completed then the employee needs to upload the task completed photo and update the task status. Employees can see address details on Google map. Also, the employee can see the current location and destination location on Google map with direction

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REFERENCES

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